

# Swayam Dhir

sdhir@mun.ca | 709 327 9933 | LinkedIn

## Summary

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Recent Computer Science graduate from Memorial University of Newfoundland with hands-on experience in full stack development, data quality assurance, and AI-assisted workflows. Passionate about building intelligent, user-focused applications and leveraging modern AI tools to accelerate software development. Actively seeking entry-level software development opportunities.

## Education

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### Bachelor of Science in Computer Science

Sept 2021 – Dec 2025

*Memorial University of Newfoundland*

- **Recent Coursework:** Algorithms to AI, Intro to Data Mining, Software Development for Mobile Devices, Evolutionary Computing, Game Programming

## Work Experience

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### Mentee – AI & Software Development Track

April 2026 – Present

*TechNest Mentorship Program*

- Participating in a structured mentorship program focused on AI and software development, receiving guidance from industry professionals on technical skills and career development.
- Engaging in hands-on learning and collaborative projects aligned with current trends in artificial intelligence and software engineering.

### Full Stack Software Developer

June 2025 – December 2025

*Memorial University of Newfoundland | DIAG Labs*

- Contributed to medical image analysis software development across web design, database systems, and image labeling tools.
- Assisted in building and maintaining a web application using Python, JavaScript, and Ruby on Rails, following team best practices including code reviews and testing.
- Collaborated within a distributed team using Git, contributing to a production-quality software development workflow.
- Developed foundational knowledge of high-performance computing in the context of AI and machine learning research.

### Data Quality Assurance Analyst

June 2025 – December 2025

*Memorial University of Newfoundland | DIAG Labs*

- Conducted data profiling, cleansing, and quality control to ensure high standards of data integrity and accuracy.
- Examined medical images to identify anatomical inconsistencies and anomalies, correcting inaccurate entries to improve data quality for downstream applications.
- Produced statistics and reports to track data quality metrics and identify areas for improvement.

# Projects

## NutriPlan *(In Progress)*

- Developing an AI-powered meal planning web application using Vercel and Cursor, integrating the Spoonacular API to deliver personalized meal recommendations.
- Designed an input-driven experience where users specify dietary restrictions and weekly budget to receive a tailored grocery shopping list, meal plan, and step-by-step recipes.
- Leveraging AI tooling throughout the development workflow to accelerate feature development and iterate on UI/UX design.

## Deadlands

- Developed a top-down zombie shooter game using C++ and SFML, implementing an Entity-Component-System (ECS) architecture for modularity, performance, and scalable game object management.
- Implemented a progression saving system that tracks completed levels and unlocks subsequent ones, utilizing file I/O with a custom load method to persist and restore game state from saved `.txt` files.
- Designed and implemented a custom in-built level editor featuring a rich library of preloaded assets and animation systems, enabling intuitive level creation through asset selection and placement. Utilized ImGui to develop a user-friendly graphical interface for asset browsing and real-time entity behavior customization.
- Developed an AI system for enemy behavior combining A\* pathfinding and ray casting for intelligent navigation and line-of-sight detection.
- Utilized shaders to implement dynamic visual effects, including a day-night cycle and animated flame effects, enhancing the game's atmosphere.

## Maple Buy

- Developed a Flutter-based mobile application enabling users to discover Canadian-made alternatives to foreign grocery products through search and categorized listings.
- Implemented scalable navigation architecture using GoRouter, structuring multi-screen routing with clean separation of UI and route logic.
- Integrated a third-party Fast Foods API to fetch grocery product data and implemented asynchronous data handling for smooth UI updates.
- Utilized Firebase (Firestore & Authentication) to manage product listings, user-generated content, and real-time reviews.

## Technologies

|                 |                                                          |
|-----------------|----------------------------------------------------------|
| AI Tools        | Cursor, GitHub Copilot, Claude, ChatGPT, Vercel AI SDK   |
| Languages       | C++, C, Java, Python, JavaScript, TypeScript, Ruby, BASH |
| Web Development | HTML, CSS, React.js, Node.js, Next.js, Ruby on Rails     |
| Databases       | MongoDB, MySQL, Shelve                                   |
| Frameworks      | Flask, Bottle, Flutter, Docker, Apache Spark             |
| AI Algorithms   | A*, BFS, DFS, Reinforcement Learning, Genetic Algorithms |
| Tools           | Git, SVN, Vercel, Visual Studio, Android Studio, Postman |